Step-by-Step Guide: Installing R and RStudio

Introduction

This guide provides step-by-step instructions for installing **R** (the statistical programming language) and **RStudio** (an integrated development environment for R) on Windows, macOS, or Linux using Overleaf-compatible LaTeX formatting.

1 Step 1: Install R

Windows / macOS

1. Visit the Comprehensive R Archive Network (CRAN):

   `https://cran.r-project.org/`

2. Select your operating system:
   - Windows: Click `Download R for Windows`
   - macOS: Click `Download R for macOS`
   - Linux: Choose your distribution
3. Download the latest stable version.
4. Open the installer file.
5. Follow installation prompts:
   - Accept default settings unless customization is needed.
   - Recommended: Keep default installation path.
6. Complete installation.

Verification

After installation:

1. Open R.
2. Confirm it launches successfully.
3. You should see the R Console.

2 Step 2: Install RStudio

1. Visit the Posit (formerly RStudio) download page:

   https://posit.co/download/rstudio-desktop/

2. Download **RStudio Desktop Free**.
3. Choose the correct installer for your operating system.
4. Open the installer.
5. Follow setup instructions:
   - Accept defaults.
   - Ensure R is already installed before opening RStudio.
6. Finish installation.

3 Step 3: Open RStudio

1. Launch RStudio.
2. RStudio should automatically detect your R installation.
3. Verify the following panes appear:
   - Console
   - Environment
   - Files/Plots/Packages
   - Script Editor

4 Step 4: Install Essential Packages

In the RStudio Console, run:

```
install.packages(c("tidyverse", "ggplot2", "dplyr", "rmarkdown"))
```

These packages are commonly used for:

- Data manipulation
- Visualization
- Statistical modelling
- Report generation

5 Step 5: Test Installation

Run the following code:

```
x <- c(1,2,3,4,5)
mean(x)
```

Expected output:

```
[1] 3
```

6 Troubleshooting

- If RStudio cannot find R:
  - Reinstall R first
  - Restart computer
  - Reopen RStudio
- If package installation fails:
  - Check internet connection
  - Update R version
  - Choose a CRAN mirror manually

7 Recommended Next Steps

- Learn basic R syntax
- Practice data import/export
- Explore `ggplot2` for visualization
- Learn `R Markdown` for reproducible reports
- Use GitHub for version control

Conclusion

You are now ready to begin statistical programming, data analysis, and reproducible research using R and RStudio.